B Mathematical Implementation

B.1 The Bellman Equation in Frequency Space

Standard distributional Bellman equation:

$$V^\pi(x) \stackrel{D}{=} R + \gamma V^\pi(X')$$

In the probability domain, this requires a convolution of their probability density functions (PDFs):

$$P_V(x) = (P_R * P_{\gamma V'})(x)$$

We define the Characteristic Function (CF) of a random variable, which represents the PDF in frequency space using the Fourier transform:

$$C_V(\omega) \triangleq \mathbb{E}\left[e^{i\omega G}\right]$$

Applying this to our Bellman equation, the convolution becomes an element-wise multiplication:

$$C_V(\omega) = \mathbb{E}\left[e^{i\omega(R+\gamma V^\pi(X'))}\right] = \mathbb{E}\left[e^{i\omega R}\right] \cdot \mathbb{E}\left[e^{i\omega\gamma V^\pi(X')}\right]$$

This gives us the Bellman equation in frequency space:

$$\tilde{V}^\pi(\omega; x) = \mathbb{E}\left[e^{i\omega R}\tilde{V}^\pi(\gamma\omega; X')\right]$$

B.2 Practical Architecture

To address the computational bottleneck of evaluating the characteristic function over the entire continuous real line $\omega \in (-\infty, \infty)$, we apply domain truncation and discretization. First, we restrict the evaluation to a bounded frequency range $[-W, W]$. Next, we discretize this continuous interval into a finite ensemble of K points, forming the discrete grid $\Omega = \{\omega_1, \omega_2, \ldots, \omega_K\}$. By replacing the infinite continuous spectrum with this finite K-dimensional vector representation, the evaluation of the Bellman operator becomes computationally tractable for neural network architectures.

If we model the neural network $\tilde{V}_\theta$ to take both ω and x as arguments, we would need to perform K forward passes for each $\omega \in \Omega$, which creates a significant bottleneck in training. Instead, the network takes only x as input and outputs a vector corresponding to a discrete set of frequencies:

$$\tilde{V}_\theta(x) \to \begin{bmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_K \end{bmatrix}$$

This allows the loss calculation to require only a single forward pass per transition sample.

B.3 Loss Calculation

The target is defined as $\tilde{T}\tilde{V}(x')$. The loss is integrated over our discrete frequency range Ω with a weighting function $w(\omega)$:

$$\mathcal{L}(x, x', \theta, \theta', \omega) = \int_{\omega\in\Omega} \left| e^{i\omega R}\tilde{V}_{\theta'}(\gamma\omega; x') - \tilde{V}_\theta(\omega; x) \right|^2 w(\omega) d\omega$$

$$\mathcal{L}(x, x', \theta, \theta', \omega) = \sum_{\omega\in\Omega} \left| e^{i\omega R}\tilde{V}_{\theta'}(\gamma\omega; x') - \tilde{V}_\theta(\omega; x) \right|^2 w(\omega)$$

B.4 Collapse Function

To extract the expected scalar value (such as the standard Value or Q-function) from the CVF, we evaluate the first derivative of the characteristic function, $\frac{d}{d\omega}\tilde{V}(\omega)$, at $\omega = 0$.

Taking the derivative of the CF with respect to ω:

$$\frac{d}{d\omega}\tilde{V}(\omega) = \frac{d}{d\omega}\mathbb{E}\left[e^{i\omega G}\right] = \mathbb{E}\left[\frac{d}{d\omega}e^{i\omega G}\right] = \mathbb{E}\left[iGe^{i\omega G}\right]$$

Evaluating this at $\omega = 0$:

$$\frac{d}{d\omega}\tilde{V}^{\pi}(\omega; x)\Big|_{\omega=0} = \mathbb{E}\left[iG^{\pi}(x)e^{0}|X = x\right] = i\mathbb{E}\left[G^{\pi}(x)|X = x\right]$$

To isolate the expected return, we multiply the derivative by $-i$. Since the derivative evaluates to a purely imaginary term, this operation algebraically cancels the imaginary unit (as $-i^2 = -(-1) = 1$):

$$V^{\pi}(x) = -i\frac{d}{d\omega}\tilde{V}^{\pi}(\omega; x)\Big|_{\omega=0} = -i\left(i\mathbb{E}\left[G^{\pi}(x)|X = x\right]\right) = -i^2\mathbb{E}\left[G^{\pi}(x)|X = x\right] = \mathbb{E}\left[G^{\pi}(x)|X = x\right]$$

This means that by evaluating the derivative of the characteristic function at frequency zero and multiplying the result by $-i$, we effectively isolate the expected return, recovering the standard Value function for a specific state x.

B.5 Polar Representation

We can express the complex CVF in polar coordinates using magnitude $M(\omega)$ and phase $\Theta(\omega)$:

$$\tilde{V}(\omega) = M(\omega)e^{i\Theta(\omega)}$$

By definition, a valid CF must satisfy two conditions at $\omega = 0$: $M(0) = 1$ and $\Theta(0) = 0$.

Taking the derivative using the product rule:

$$\frac{d\tilde{V}(\omega)}{d\omega} = \frac{dM(\omega)}{d\omega}e^{i\Theta(\omega)} + M(\omega)\cdot i\frac{d\Theta(\omega)}{d\omega}e^{i\Theta(\omega)}$$

Evaluating at $\omega = 0$:

$$\frac{d\tilde{V}(\omega)}{d\omega}\Big|_{\omega=0} = M'(0) + i\Theta'(0)$$

Linking this back to our expected return equation:

$$-i\left[M'(0) + i\Theta'(0)\right] = \Theta'(0) - iM'(0)$$

Since the expected return $\mathbb{E}[G]$ must be a real number, we take the real part:

$$\mathbb{E}[G] = \text{Re}\left[\Theta'(0) - iM'(0)\right] = \Theta'(0)$$

B.6 Numerical Implementations of the Collapse Function

To compute $\Theta'(0)$ numerically, we consider two approaches.

B.6.1 Finite Difference (approximation of the first derivative)

$$\frac{d\Theta}{d\omega} \approx \frac{\Theta(\omega) - \Theta(0)}{\omega}$$

However, this relies on a single point of failure near zero, expanding entirely on the precision of a single forward pass, which will collapse.

B.7 Ordinary Least Squares (Gaussian Assumption)

If we assume the return distribution is locally Gaussian $X \sim \mathcal{N}(\mu, \sigma^2)$, its CF is:

$$C_X(\omega) = e^{i\mu\omega - \frac{1}{2}\sigma^2\omega^2}$$

Here, the phase is purely linear with respect to the mean:

$$\Theta(\omega) = \mu\omega$$

Taking the derivative with respect to ω directly yields the mean:

$$\frac{d\Theta}{d\omega} = \mu$$

Because this relationship is strictly linear and passes through the origin, we can algebraically isolate μ at any non-zero frequency as:

$$\mu = \frac{\Theta(\omega)}{\omega}$$

To increase robustness against noise from the neural network outputs, we use an Ordinary Least Squares (OLS) regression over multiple symmetric frequency pairs near the origin rather than relying on a single point:

$$Q = \mu = \frac{\sum_{\omega \in \Omega} \omega \cdot \Theta(\omega)}{\sum_{\omega \in \Omega} \omega^2}$$

This results in a single, stable scalar Q-value.

B.8 Conclusion

To successfully train and deploy CVI as an algorithm, we must extract a scalar representation (the Q-value) from the distributional return to perform greedy action selection. Collapsing this distribution into a scalar necessitates extracting the first moment by taking the derivative of the complex CVF at frequency $\omega = 0$. By representing the CVF in polar coordinates, we demonstrate that evaluating this derivative simplifies to finding the derivative of the phase at frequency zero. Finally, driven by the need for numerical stability in neural network approximations, we are currently exploring the use of Ordinary Least Squares (OLS) regression over symmetric frequency pairs to robustly estimate this phase slope.